

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	MURALI KRISHNAN N	Reg. No.:	192524090
Date:	22/08/26	Duration:	60 minutes

Code of Conduct

I MURALI KRISHNAN N (192524090)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:

Annexure A – Research Paper Review / Literature Survey

to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers (25%)	/5
Comparative Analysis (20%)	/5
Critical Analysis & Research Gap (25%)	/5
Future Directions & Relevance (15%)	/5
Documentation, Citations & References (15%)	/5
TOTAL	/25

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Abstract

Microgrids that combine solar photovoltaic (PV) generation, wind generation, battery energy storage systems (BESS), and utility-grid interconnection are increasingly deployed in campuses, industrial parks, and residential communities to improve energy resilience and to accelerate the integration of renewable energy. Conventional, rule-based energy management systems (EMS) struggle to cope with the intermittent nature of renewable sources and with fast-changing load profiles, which leads to inefficient energy utilisation, poor load balancing, and higher operating costs. Artificial Intelligence (AI) — spanning classical machine learning, deep learning, deep reinforcement learning (DRL) and metaheuristic optimisation — has emerged as the dominant approach to overcome these limitations by enabling accurate demand and generation forecasting, adaptive battery dispatch, automated load balancing, and real-time operational visibility. This report presents a structured literature survey of ten recent, peer-reviewed research papers relevant to the assigned problem statement, "AI-Based Smart Microgrid Energy Management and Load Balancing Platform." Each paper is individually reviewed for its objectives, methodology, tools/datasets, and key findings; the papers are then compared using a common set of parameters, and the limitations and open research gaps of the existing literature are analysed. Building on the identified gaps, the report proposes a conceptual architecture for an AI-powered microgrid management platform and outlines practical future research directions, including edge-AI forecasting, hybrid RL-metaheuristic dispatch, explainable AI (XAI), and standardised multi-microgrid benchmarking.

1. Introduction

1.1 Industry Context

Microgrids are increasingly adopted in campuses, industries, and residential communities to improve energy resilience and to integrate renewable energy sources at scale. The convergence of AI, the Internet of Things (IoT), smart meters, and cloud computing now enables intelligent energy management, real-time monitoring, and demand-response optimisation that were not possible with legacy supervisory control and data acquisition (SCADA)-only systems. As distributed energy resources (DERs) such as rooftop solar, small wind turbines, and battery storage proliferate, the operational complexity of balancing generation, storage, and consumption in real time has grown substantially, motivating a shift from static, rule-based control to data-driven, AI-based control.

1.2 Problem Statement

Microgrids must balance energy generation, storage, and consumption while accommodating intermittent renewable energy sources. Conventional energy management systems often lack predictive capabilities, resulting in inefficient energy utilisation and higher operational costs. An AI-powered microgrid management platform should monitor electricity generation and consumption, forecast energy demand, optimise battery storage utilisation, balance electrical loads, and provide real-time operational dashboards. The system should improve energy reliability, reduce operational expenses, and support sustainable power management.

1.3 Objectives

- Monitor energy generation and consumption in real time.
- Forecast electricity demand and renewable generation using AI.
- Optimise battery storage utilisation (charge/discharge scheduling).
- Balance electrical loads automatically across sources.
- Integrate renewable energy sources (solar and wind) efficiently.
- Generate energy performance reports and dashboards.
- Reduce operational costs of microgrid operation.
- Improve overall microgrid reliability and resilience.

1.4 Expected Outcomes

- Improved energy efficiency and demand-supply balance.
- Better automated load balancing across connected feeders.
- Reduced electricity and operational costs.
- Increased utilisation of renewable energy generation.

- Enhanced grid reliability and fault resilience.
- Sustainable, data-driven energy planning and reporting.
- Reduced carbon emissions through optimised dispatch.

1.5 Purpose of This Survey

The purpose of this literature survey is to study existing AI-based approaches to microgrid energy management and load balancing, to compare their methodologies, datasets/tools, and reported performance, to identify the limitations and research gaps in the current body of work, and to propose future research directions and a conceptual system architecture that directly addresses the assigned problem statement.

2. Literature Survey

Ten recent research papers (2020-2026) relevant to AI-based microgrid energy management and load balancing were identified from reputed sources including Scientific Reports (Nature), MDPI (Energies, Processes), IEEE Access, ScienceDirect (Journal of Energy Storage), E3S Web of Conferences, The Journal of Engineering (IET/Wiley), and arXiv. Each paper is reviewed below in terms of its objectives, methodology, tools/datasets used, and major findings.

2.1 AI-Driven Energy Management Systems for Microgrids: Optimizing Renewable Energy Integration and Load Balancing

Bhardwaj, D.; Shalini, M.; Kanmani Jebaseeli, S.; Jadhav, S.; Alabdeli, H.; Sutar, V.; Senthil Kumar, R. (2024), E3S Web of Conferences, Vol. 591, Article 01005.

Objective: to design an AI-driven Energy Management System (EMS) that jointly optimises power distribution and load balancing in renewable-integrated microgrids. Methodology: the authors develop a machine-learning-based EMS that forecasts energy demand and adapts power allocation in real time across two interconnected microgrids sharing surplus/deficit energy. Tools/Dataset: simulated multi-microgrid environment with PV generation, load-demand time series, and battery state-of-charge (SoC) modelling. Key Findings: the proposed AI-driven EMS shows significant improvement in energy efficiency and grid stability compared with traditional (non-AI) EMS approaches, and demonstrates that energy sharing between interconnected microgrids improves overall utilisation of surplus renewable generation.

2.2 Machine Learning-Based Energy Management and Power Forecasting in Grid-Connected Microgrids with Multiple Distributed Energy Sources

Singh, A.R.; Kumar, R.S.; Bajaj, M.; Khadse, C.B.; Zaitsev, I. (2024), Scientific Reports, 14, Article 19207.

Objective: to improve short-term power-generation forecasting for grid-connected microgrids with multiple distributed energy sources (MDES). Methodology: a Support Vector Regression (SVR) model is trained on historical energy-production data, weather patterns, and grid conditions to forecast solar PV and wind power

output. Tools/Dataset: real-world solar and wind generation datasets with weather covariates; SVR is benchmarked against linear regression. Key Findings: the SVR model achieves a Mean Squared Error of 2.002 (solar PV) and 3.059 (wind), clearly outperforming linear regression; the improved forecasts enable adaptive real-time energy management that delivers a 10% improvement in supply-demand balance, a 15% reduction in peak load demand, and a 12% increase in renewable energy utilisation.

2.3 A Comprehensive Review of Microgrid Energy Management Strategies Considering Electric Vehicles, Energy Storage Systems, and AI Techniques

Khan, M.R.; Haider, Z.M.; Malik, F.H.; Almasoudi, F.M.; Alatawi, K.S.S.; Bhutta, M.S. (2024), *Processes*, 12(2), Article 270 (MDPI).

Objective: to comprehensively review microgrid EMS strategies that account for electric vehicles (EVs), energy storage systems (ESS), and AI-based optimisation techniques. Methodology: a structured literature review comparing demand-side management strategies, optimisation algorithms, and control architectures. Tools/Dataset: synthesis of over 100 published studies (no primary dataset). Key Findings: the review highlights the growing role of AI in strengthening smart-grid technologies and in handling load-management uncertainty, discusses vehicle-to-grid (V2G) integration as a flexible storage resource, and identifies cybersecurity of AI-controlled microgrids as an emerging concern that current EMS designs rarely address.

2.4 A Smart Microgrid System with Artificial Intelligence for Power-Sharing and Power Quality Improvement

Nair, D.R.; Nair, M.G.; Thakur, T. (2022), *Energies*, 15(15), Article 5409 (MDPI).

Objective: to improve power-sharing and power-quality in grid-integrated smart microgrid systems using an AI-based control algorithm. Methodology: an artificial-intelligence-based Icos-phi control algorithm is designed to manage power sharing under load variation, battery state-of-charge, and power-tariff uncertainty; validated through simulation and a hardware prototype. Tools/Dataset: hardware-in-the-loop prototype with nonlinear load; simulation of multiple smart microgrids coupled with renewable sources. Key Findings: the AI-based controller improves power quality (harmonic mitigation, reactive-power compensation) and enables adaptive power-sharing at the point of common coupling; economic viability of the proposed controller design was also confirmed.

2.5 Energy Management System for Multi-Interconnected Microgrids during Grid-Connected and Autonomous Operation Modes Considering Load Management

Shaker, H.K.; Keshta, H.E.; Mosa, M.A.; Ali, A.A. (2024), *Scientific Reports*, 14, Article 24359.

Objective: to develop an AI-based economic-dispatch and load-management approach for three interconnected microgrids operating in both grid-connected and autonomous (islanded) modes. Methodology: a day-ahead scheduling method computes optimal set points for each energy source, while a metaheuristic One-to-One-Based Optimizer (OOBO) solves the load-shifting and economic-dispatch optimisation problem. Tools/Dataset:

simulated three-microgrid test system with controllable and non-controllable loads. Key Findings: the OOB0-based strategy reduces daily operating cost by 1.6% in grid-connected mode and 0.47% in islanded mode compared with baseline dispatch, while maintaining safe operating constraints across all three microgrids.

2.6 Multi-Objective Optimization Algorithms for Energy Management Systems in Microgrids: A Control Strategy Based on a PHIL System

Islam, S.; Mostaghim, S.; Hartmann, M. (2025), arXiv:2505.18210 [eess.SY].

Objective: to propose an adaptive, real-time multi-objective optimisation strategy for microgrid EMS that jointly balances fuel consumption, load mismatch, power quality, battery degradation, and renewable utilisation. Methodology: a weighted, preference-based multi-objective optimiser is integrated with a Power-Hardware-In-the-Loop (PHIL) test-bed comprising PV, a grid-tied inverter, battery storage, the utility grid, and a diesel generator; the optimiser dynamically updates system states using real-time experimental data. Tools/Dataset: real-time PHIL experimental platform (not a purely simulated dataset). Key Findings: the adaptive multi-objective approach maximises PV utilisation while minimising power mismatch and stabilising battery SoC across varying operating conditions, and outperforms a non-optimised baseline system in practical, hardware-validated tests.

2.7 Adaptive Multi-Objective Optimization of Microgrid Energy Management Using Deep Reinforcement Learning Considering Battery Degradation and Renewable Uncertainty

Altmanian, M.R.M.; Basem, A.; Saydullaev, B.; Atamuratova, Z.; Madaminov, B.; Umarov, A.; Benti, N.E.; Chaka, M.D. (2026), Scientific Reports (Springer Nature).

Objective: to develop a Deep Reinforcement Learning (DRL) based EMS that optimises microgrid operation while explicitly considering operational cost, battery degradation, and renewable-generation uncertainty. Methodology: a Deep Q-Network (DQN) agent is trained to manage energy flows in a simulated microgrid (solar PV, battery storage, controllable loads, grid connection); the reward function combines operating cost, battery degradation, and renewable-utilisation objectives. Tools/Dataset: simulated microgrid environment benchmarked against a Model Predictive Control (MPC) baseline. Key Findings: the DRL-based EMS achieves a 12.01% reduction in operational cost, an 8.19% reduction in battery degradation, and a 10.39% increase in renewable-energy utilisation relative to the MPC benchmark, and maintains robust performance (only an 8.9% cost increase versus 21.5% for conventional methods) under severe forecast-error conditions.

2.8 Deep Reinforcement Learning-Based Control Strategy for Integration of a Hybrid Energy Storage System in Microgrids

Kumar, K.; Kwon, S.; Bae, S. (2025), Journal of Energy Storage, 108, Article 114936 (ScienceDirect).

Objective: to design a DRL-based control strategy for power management in microgrids that use a hybrid energy storage system (HESS) combining supercapacitors, batteries, and hydrogen storage. Methodology: a two-layer DRL control strategy is proposed — a supervisory layer that optimally distributes power among HESS

components, and a local layer that controls the switching of power-electronic converters. Tools/Dataset: simulation of a DC microgrid under multiple operating scenarios with fluctuating renewable generation and load. Key Findings: the two-layer DRL strategy achieves optimal power-sharing among heterogeneous storage components, improves DC-bus voltage regulation, and effectively addresses rapid, uncertain load and generation variations that conventional rule-based HESS controllers cannot handle.

2.9 Microgrid Energy Management System with Embedded Deep Learning Forecaster and Combined Optimizer

Suresh, V.; Janik, P.; Guerrero, J.M.; Leonowicz, Z.; Sikorski, T. (2020), IEEE Access, 8, pp. 202225-202239.

Objective: to build a microgrid EMS for a university campus that embeds a deep-learning forecaster together with a combined optimiser and an EV-charging load. Methodology: an LSTM-Autoencoder deep-learning architecture forecasts load and generation; forecasts feed a combined optimisation module that schedules dispatch and an optimised EV charging station. Tools/Dataset: real operational data from the Wroclaw University of Science and Technology campus microgrid. Key Findings: the LSTM-Autoencoder forecaster provides stable, reliable forecasting performance across repeated runs, and the statistical analysis of running time and reliability builds a practical case for adopting deep-learning forecasters in real, deployed campus-scale microgrid EMS.

2.10 DeepEMS: Multimodal Optimal Energy Management of Microgrid Systems Based on a Hybrid Multi-Stage Machine Learning Model

Safari, A.; Hashemzadeh, F.; Zare, K. (2024), The Journal of Engineering, 2024(10) (IET/Wiley).

Objective: to jointly optimise the utilisation of multiple energy sources (solar PV and storage) in a microgrid using a hybrid, multi-stage machine-learning pipeline. Methodology: DeepEMS combines several intelligent algorithms in stages — forecasting, classification, and optimisation — to manage dispatch decisions under uncertainty. Tools/Dataset: simulated microgrid dataset incorporating solar generation and storage-state variables. Key Findings: the multi-stage hybrid ML approach improves grid reliability and resilience to unexpected demand variation compared with single-model (non-hybrid) EMS baselines, supporting the case for combining forecasting and optimisation stages rather than relying on a single AI technique.

Fig. 2: Distribution of AI/ML Techniques Across the 10 Reviewed Research Papers

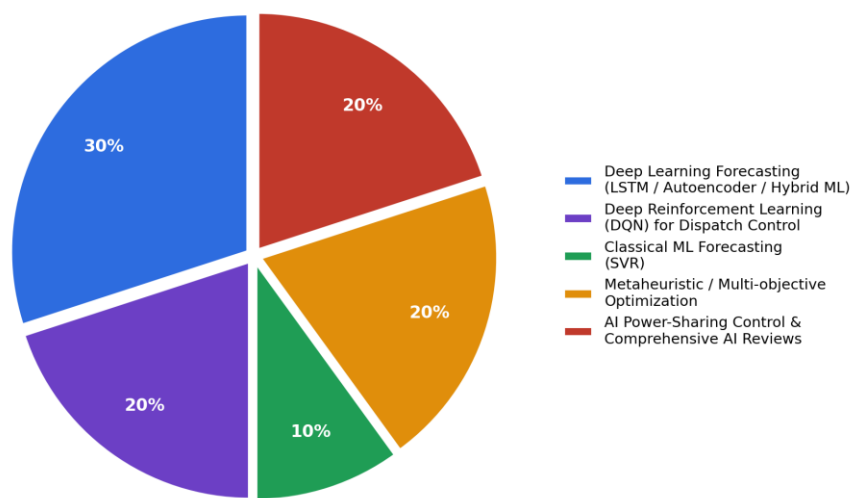


Figure 3: Distribution of AI/ML techniques across the ten reviewed research papers.

3. Comparative Analysis of Reviewed Literature

Table 1 compares the ten reviewed studies using seven common parameters: author/year, research problem, methodology, dataset/tools, key findings, and the associated limitation or research gap. The comparison shows a clear methodological split between forecasting-oriented studies (classical ML and deep learning), control/dispatch-oriented studies (deep reinforcement learning and metaheuristic optimisation), and broader AI-integration reviews.

Table 1: Comparative Literature Survey Table

#	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
1	Bhardwaj et al., 2024	Load balancing across interconnected microgrids	ML-based EMS, real-time power allocation	Simulated multi-microgrid, PV + battery	Improved efficiency & stability vs. traditional EMS	No field/hardware validation; single-site simulation only
2	Singh et al., 2024	Power-generation forecasting for MDES microgrids	Support Vector Regression (SVR)	Real solar/wind data + weather covariates	10% supply-demand balance, 15% peak-load	Classical ML; no online/continual re-training discussed

#	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
					cut, 12% RE increase	
3	Khan et al., 2024	EV, ESS and AI integration in microgrid EMS	Structured literature review	100+ synthesised published studies	Highlights AI's role vs. uncertainty; flags cybersecurity gap	Review only — no new experimental EMS validation
4	Nair et al., 2022	Power-sharing & power-quality at point of common coupling	AI-based Icosphi control algorithm	Hardware prototype + simulation	Improved power quality, adaptive power-sharing, proven economics	Small-scale hardware test; scalability to large MG clusters untested
5	Shaker et al., 2024	Economic dispatch & load shift, 3 interconnected MGs	Meta-heuristic One-to-One-Based Optimizer (OOBO)	Simulated 3-microgrid test system	Cost cut by 1.6% (grid-connected), 0.47% (islanded)	Meta-heuristic converges slowly at larger scale; no forecasting module
6	Islam et al., 2025	Real-time multi-objective dispatch under hardware constraints	Adaptive weighted multi-objective optimisation	Power-Hardware-in-the-Loop (PHIL) test-bed	Maximised PV use, stabilised battery SoC, hardware-validated	High computational/hardware cost; limited to test-bed scale
7	Altmanian et al., 2026	Cost, battery-degradation & renewable-uncertainty trade-off	Deep Reinforcement Learning (DQN)	Simulated MG vs. MPC benchmark	-12.01% cost, -8.19% battery degradation, +10.39% RE use	DRL training cost is high; reward design not generalised across MG types

#	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
8	Kumar et al., 2025	Power-sharing in hybrid storage (battery+SC+H2) microgrids	Two-layer Deep Reinforcement Learning	Simulated DC microgrid, multiple scenarios	Optimal power-sharing; improved DC-bus voltage regulation	Complex hardware assumed; limited real-world deployment data
9	Suresh et al., 2020	Campus EMS: forecasting + optimised dispatch + EV charging	LSTM-Autoencoder forecaster + combined optimiser	Real campus microgrid data (Wroclaw Univ.)	Stable, reliable multi-run forecasting performance	Single-campus dataset; generalisability to other climates unverified
10	Safari et al., 2024	Multimodal dispatch optimisation under uncertainty	Hybrid multi-stage ML pipeline (DeepEMS)	Simulated PV + storage dataset	Improved reliability & resilience vs. single-model baselines	Multi-stage pipeline adds latency; not tested for cyber-resilience

Fig. 3: Quantitative Improvements Reported by Selected AI-Based Microgrid EMS Studies

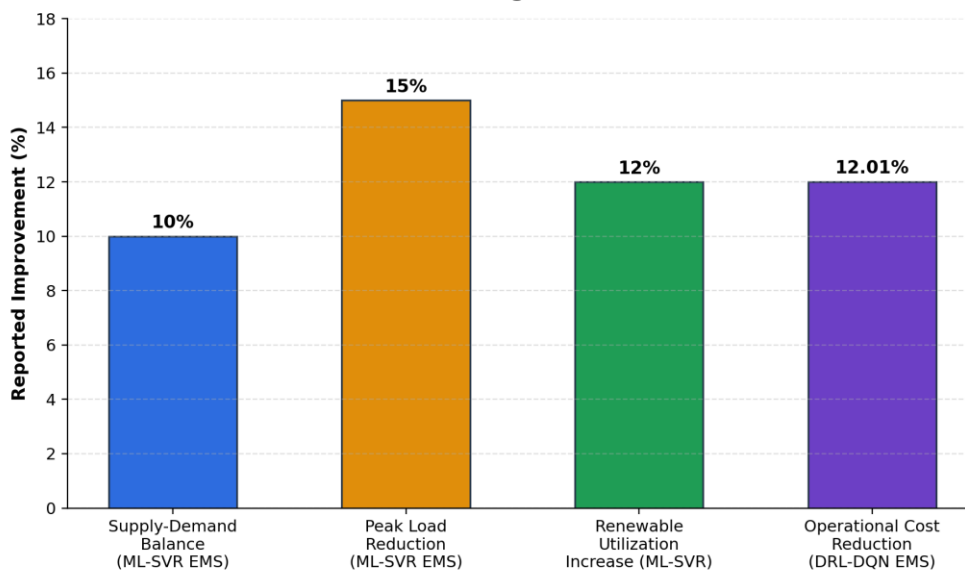


Figure 4: Quantitative improvements reported by selected AI-based microgrid EMS studies (Singh et al., 2024; Altimania et al., 2026).

4. Critical Analysis and Research Gaps

Reviewing the ten studies together reveals several recurring limitations in the current state of AI-based microgrid energy management research:

- **Limited real-time adaptability:** several EMS designs (e.g., Bhardwaj et al., 2024; Khan et al., 2024) are validated primarily in simulation and rely on periodic re-training, which limits responsiveness to sudden weather-driven generation swings or unplanned load spikes.
- **High computational cost of advanced methods:** DRL-based dispatch (Altimania et al., 2026; Kumar et al., 2025) and PHIL-based multi-objective optimisation (Islam et al., 2025) deliver strong performance but require significant training time, computing resources, or specialised hardware, which constrains deployment on low-cost edge controllers used in small campus or residential microgrids.
- **Lack of explainability and cybersecurity focus:** Khan et al. (2024) explicitly flag cybersecurity as an under-addressed concern, and none of the reviewed DRL/optimisation papers report explainable-AI (XAI) mechanisms that would let operators audit automated dispatch decisions.
- **Narrow validation scope:** most studies (Singh et al., 2024; Suresh et al., 2020; Safari et al., 2024) are validated on a single site, dataset, or climate profile, so generalisability of forecasting accuracy and control performance across different microgrid sizes, geographies, and load types remains largely unverified.
- **Few standardised, multi-microgrid benchmarks:** only Bhardwaj et al. (2024) and Shaker et al. (2024) study interconnected/multi-microgrid coordination, and each uses its own bespoke simulation setup, making cross-paper performance comparison difficult.

These gaps collectively define the research opportunity for the assigned problem statement: an AI-based smart microgrid platform that is not only accurate in forecasting and optimal in dispatch, but also computationally lightweight, explainable, secure, and validated across realistic, multi-microgrid operating conditions.

5. Proposed System and Future Research Directions

5.1 Conceptual Architecture

Building on the objectives of the assigned problem statement and on the gaps identified in Section 4, this report proposes a layered architecture for an AI-based smart microgrid energy management and load-balancing platform (Figure 1). Solar PV, wind generation, battery storage, and the utility grid connection feed data through an IoT sensor and smart-meter layer into a cloud data-acquisition and communication layer. A central AI/ML engine performs three coordinated functions: (i) short-term load and generation forecasting using deep-learning models such as LSTM, informed by the forecasting approaches reviewed in Section 2 (Singh et al., 2024; Suresh

et al., 2020); (ii) dispatch and battery-storage optimisation using deep reinforcement learning, following the DQN-based approach of Altimania et al. (2026) and the two-layer DRL strategy of Kumar et al. (2025); and (iii) an automated load-balancing controller inspired by the AI-based power-sharing algorithm of Nair et al. (2022). The outputs drive automated load balancing and scheduling, optimised battery charge/discharge, and a real-time operational dashboard that reports performance to campus, industrial, or residential facility operators.

Fig. 1: Architecture of an AI-Based Smart Microgrid Energy Management and Load Balancing Platform

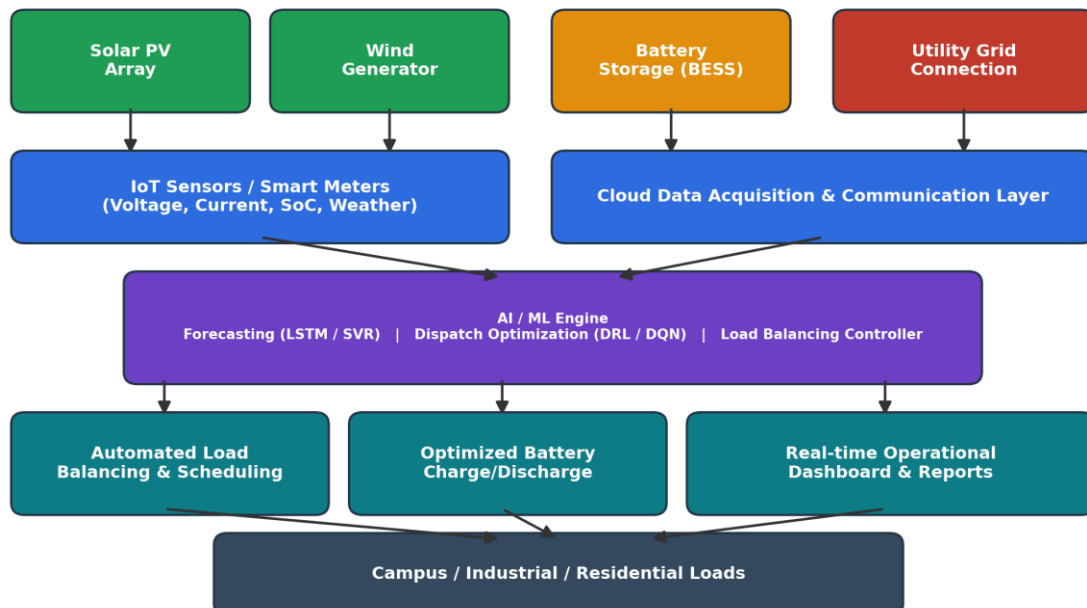


Figure 1: Proposed architecture of the AI-based smart microgrid energy management and load-balancing platform.

5.2 Alignment with Assigned Objectives

The proposed architecture directly maps onto the eight objectives stated in Section 1.3: the IoT/smart-meter layer supports real-time monitoring; the AI/ML forecasting module addresses demand forecasting; the DRL-based dispatch module optimises battery storage utilisation and integrates renewable sources; the load-balancing controller automatically balances electrical loads; and the operational dashboard generates the energy performance reports needed to reduce operational costs and improve reliability.

5.3 Identified Research Gaps and Future Directions

Figure 2 summarises the four principal research gaps identified in Section 4 and links each to a corresponding, actionable future research direction for the assigned problem.

Fig. 4: Identified Research Gaps and Corresponding Future Research Directions

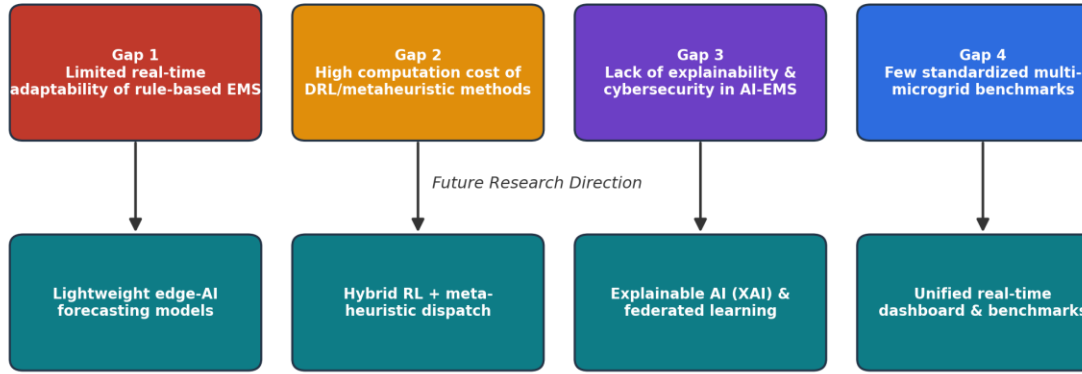


Figure 2: Identified research gaps and corresponding future research directions.

- **Lightweight edge-AI forecasting:** compress LSTM/DRL forecasting models (e.g., via quantisation or knowledge distillation) so that they can run on low-cost edge gateways rather than requiring continuous cloud connectivity, addressing the real-time adaptability gap.
- **Hybrid RL + metaheuristic dispatch:** combine the fast, learned policies of DRL (Altıman et al., 2026; Kumar et al., 2025) with the constraint-handling strength of metaheuristics such as OBO (Shaker et al., 2024) to reduce training cost while retaining near-optimal dispatch quality.
- **Explainable AI (XAI) and federated learning:** incorporate XAI techniques so operators can audit automated load-balancing and battery-dispatch decisions, and use federated learning across sites to improve forecasting accuracy without centralising sensitive consumption data, partially addressing the cybersecurity gap noted by Khan et al. (2024).
- **Unified real-time dashboard and multi-microgrid benchmarks:** extend single-site dashboards toward a standardised benchmark suite for interconnected, multi-microgrid coordination, building on the early multi-microgrid work of Bhardwaj et al. (2024) and Shaker et al. (2024), so that future AI-EMS studies can be compared on a common basis.

6. Conclusion

This literature survey reviewed ten recent, peer-reviewed studies on AI-based microgrid energy management and load balancing, spanning classical machine learning (SVR), deep-learning forecasting (LSTM/Autoencoder), deep reinforcement learning (DQN-based dispatch), metaheuristic multi-objective optimisation, and broader review/synthesis work. Across the reviewed literature, AI-based EMS consistently outperform traditional, rule-based EMS on key metrics such as supply-demand balance, peak-load reduction, renewable-energy utilisation, and operational cost, with reported improvements ranging from roughly 1-2% up to 15% depending on the technique and operating mode. At the same time, the comparative and critical analysis identified consistent gaps around real-time adaptability, computational cost, explainability, cybersecurity, and

the scarcity of standardised multi-microgrid benchmarks. The proposed AI-based smart microgrid energy management and load-balancing platform, built around an IoT/cloud sensing layer, a hybrid forecasting-plus-dispatch AI engine, and a real-time operational dashboard, directly targets the assigned problem statement and offers a practical path toward more reliable, cost-effective, and sustainable microgrid operation. Future work following the directions outlined in Section 5 — particularly lightweight edge-AI forecasting and explainable, hybrid RL-metaheuristic dispatch — would further close the gap between promising simulation results and dependable, real-world deployment.

7. References

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